# Joe Kanjirathingal

Machine Learning | Python, PyTorch, NLP | Data Analysis
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## SUMMARY

B.Tech Artificial Intelligence student (graduating March 2027) seeking machine learning, data science and data analyst internships. Fine-tuned MiniLM and served it with ONNX, built an MPNet-embedding classifier and compared every model with a baseline; built tested data pipelines over 110M+ records with Python, SQL and DuckDB.

## EDUCATION

**B.Tech in Artificial Intelligence | Sep 2023 - Mar 2027 (expected)**
Vidya Academy of Science and Technology, Thrissur, Kerala

## TECHNICAL SKILLS

**Machine Learning:** scikit-learn, PyTorch, Hugging Face Transformers, ONNX, NLP, classification, model evaluation
**Programming and Data:** Python, SQL (window functions, CTEs), pandas, NumPy, DuckDB, ETL, data cleaning
**Statistics:** regression, hypothesis testing, time-series analysis, forecasting, cohort analysis
**BI and Reporting:** Power BI (DAX, Power Query, data modeling), Excel (PivotTables, slicers, XLOOKUP, SUMIFS), KPIs
**Visualization and Tools:** Flourish, web scraping (Scrapling), FastAPI, React, PostgreSQL, Git

## SELECTED PROJECTS

**QuoteCheck - Supplier Quotation Review and Product Matching | Project**
*Oct 2026 - Present | Python, PyTorch, FastAPI, React, PostgreSQL, ONNX, Vercel*
- Built a quotation-review app with PDF/XLSX extraction, source-linked corrections, exact pricing and exports.
- Fine-tuned MiniLM on 19,835 product pairs; achieved 74.3% validation average precision and 64.7% F1 on a 4,500-pair WDC Products benchmark, versus 43.4% F1 for the logistic baseline.
- Deployed ONNX CPU inference on Vercel with PostgreSQL; verified by 247 automated tests and 3 browser workflows.

**IntentDesk - NLP Classification and Evaluation | Live demo**
*Oct 2026 | Python, scikit-learn, PyTorch, FastAPI, SQLite*
- Built an NLP workflow for 150 intent categories using frozen pretrained MPNet embeddings, a downstream classifier and probability calibration.
- 96.24% accuracy on 4,500 CLINC150 test requests versus 91.96% for a lexical baseline; kept human review after unsupported-request detection failed.

**Kerala Vegetable Prices - Power BI Dashboard, Price Analysis and Spike Warning | Live site**
*Nov 2026 | Power BI, Excel, Python, SQL, DuckDB, scikit-learn, statsmodels, Scrapling*
- Built a 3-page Power BI dashboard on a star-schema data model (9 tables, Power Query, 20 DAX measures, KPI cards, slicers) and an Excel workbook with PivotTables whose 12 KPIs match Python.
- Cleaned 26.5M Agmarknet records (5 GB) in a tested ETL pipeline; found Kerala paid 1.9-3.3x the all-India wholesale price, with 20%+ gaps between markets 40 km apart.
- Trained a logistic regression spike-warning model on a chronological split: 0.33 PR-AUC versus 0.04 chance, warning of 4 of 5 held-out spikes a median 16 days early.

**Retail Analytics Lab - E-commerce Data Warehouse and Analysis | Live site**
*Oct 2026 | Python, SQL, DuckDB, Parquet, NumPy, SciPy, Excel*
- Built a reproducible Python and DuckDB pipeline and star schema over 110M e-commerce events (4.6 GB); 14 data tests reconcile rows and $505.1M of revenue from raw files to every output.
- Detected a 48-hour purchase-tracking outage that created a false 53% weekend revenue effect; added automated gap detection and excluded the affected days from conversion analysis.
- Wrote SQL marts (window functions, cohorts, Pareto) and a business memo: 42% of buyers drive 79% of revenue; 13-day forecast at 5.6% MAPE versus a 5.9% seasonal-naive baseline.

## TRAINING

**Python Full-Stack Development | SMEClabs, Kochi | Jun 2026 - Dec 2026 (expected)**
Six-month programme covering Python, SQL, Django, HTML/CSS and JavaScript.